

Results

About this document:

This file serves as supplementary material (<https://doi.org/10.17605/OSF.IO/CV8WB>) for the article "Best practices for your confirmatory factor analysis: A JASP and lavaan tutorial" by Rogers (2024), published in *Behavior Research Methods* (<https://doi.org/10.3758/s13428-024-02375-7>).

Software and reproducibility:

This analysis was developed in **JASP v0.17.1** (<https://github.com/jasp-stats/jasp-desktop/releases/tag/v0.17.1>). You may encounter reproducibility issues if using other versions of JASP, as the software is under active development and features may change across versions.

Data context:

This file presents confirmatory factor analysis (CFA) results for the World Health Organization Quality of Life-Abbreviated Version (WHOQOL-BREF) treating items as **ordinal variables**, consistent with best practices for Likert-type scales in applied social sciences. The analyses employ the **Diagonally Weighted Least Squares (DWLS)** estimator (with robust standard errors: **RDWLS**), also known as WLSMV (Weighted Least Squares Mean and Variance adjusted), which is appropriate for ordinal data and robust to violations of multivariate normality. The sample consists of $n = 1,047$ observations collected in Brazil. Four competing models are evaluated: (1) a four-correlated-factor model (4F-A) with all 24 items, (2) the same model excluding item Q5P (4F-B), (3) a bifactor model with general and specific factors, and (4) a second-order hierarchical model. Models are compared using fit indices (CFI, TLI, RMSEA, SRMR), modification indices, and reliability coefficients (McDonald's ω , GLB). The technical comments included throughout this document provide detailed interpretations of some important analysis output, contextualize findings within contemporary psychometric literature, and highlight methodological considerations for conducting rigorous CFA with ordinal data.

CFA_4F

Model fit

Chi-square test

Model	X²	df	p
Baseline model	71098.143	276	
Factor model	1339.798	245	< .001

The chi-square test ($\chi^2 = 1339.798$, $df = 245$, $p < .001$) indicates significant discrepancy between the observed and model-implied covariance matrices. However, chi-square is highly sensitive to sample size ($n = 1047$), commonly rejecting well-fitting models in large samples. Therefore, we should rely primarily on approximate fit indices rather than this exact-fit test. The baseline model (independence model) serves as a reference, assuming no relationships among variables.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.985
Tucker-Lewis Index (TLI)	0.983
Bentler-Bonett Non-normed Fit Index (NNFI)	0.983
Bentler-Bonett Normed Fit Index (NFI)	0.981
Parsimony Normed Fit Index (PNFI)	0.871
Bollen's Relative Fit Index (RFI)	0.979
Bollen's Incremental Fit Index (IFI)	0.985
Relative Noncentrality Index (RNI)	0.985

Information criteria

 Information criteria are only available with ML-type estimators

Akaike (AIC)

Bayesian (BIC)

Sample-size adjusted Bayesian (SSABIC)

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.065
RMSEA 90% CI lower bound	0.062
RMSEA 90% CI upper bound	0.069
RMSEA p-value	2.258×10^{-13}
Standardized root mean square residual (SRMR)	0.057
Hoelter's critical N ($\alpha = .05$)	221.561
Hoelter's critical N ($\alpha = .01$)	234.759
Goodness of fit index (GFI)	0.987
McDonald fit index (MFI)	0.593
Expected cross validation index (ECVI)	

The four-factor model demonstrates excellent fit to the data based on conventional fit indices. The CFI (0.985) and TLI (0.983) exceed the stringent threshold of 0.95, the RMSEA (0.065, 90% CI [0.062, 0.069]) falls within the acceptable range (< 0.08) though slightly above the "close fit" criterion of 0.05, and the SRMR (0.057) is below the 0.08 threshold, suggesting minimal discrepancy between observed and predicted correlations. However, **these traditional cutoff values must be interpreted with caution**. As discussed in the Appendix (Table 3, Step 4), traditional cutoffs (RMSEA $< .06$, CFI $> .95$, SRMR $< .06$) were derived from simulation studies based on maximum likelihood (ML) estimation with continuous data, typically considering models with only three factors, fifteen items, and factor loadings between 0.7 and 0.8. When using DWLS estimation for ordinal data, these indices behave differently: DWLS tends to produce lower RMSEA values and higher CFI values than ML, making traditional cutoffs potentially misleading. Furthermore, these cutoffs are affected by the number of items and factors, model degrees of freedom, magnitude of factor loadings, type of misfit, and missing data patterns. Rather than relying on arbitrary threshold values, fit indices should be interpreted as **effect size measures** that indicate the **degree of fit/misfit**, not as hypothesis tests for binary accept/reject decisions. A more appropriate approach involves using **Dynamic Fit Index (DFI)** cutoffs tailored to the specific model characteristics (see supplementary materials for DFI calculations using the R dynamic package), which accounts for the ordinal nature of the data, the number of factors and items, and the estimation method employed.

	R²
Q3_F	0.258
Q4_F	0.179
Q10_F	0.667
Q15_F	0.409
Q16_F	0.351
Q17_F	0.849
Q18_F	0.786
Q5_P	0.548
Q6_P	0.449
Q7_P	0.441
Q11_P	0.434
Q19_P	0.740
Q26_P	0.410
Q20_S	0.668
Q21_S	0.340
Q22_S	0.564
Q8_A	0.260
Q9_A	0.390
Q12_A	0.508
Q13_A	0.422
Q14_A	0.586
Q23_A	0.325
Q24_A	0.232
Q25_A	0.280

The R² values represent the proportion of variance in each observed indicator explained by its corresponding latent factor. Items Q17F (physical; R² = 0.849), Q18F (physical; R² = 0.786), Q19P (psycho; R² = 0.740), and Q10F (physical; R² = 0.667) exhibit the highest explained variance, indicating strong relationships with their respective factors. Conversely, Q4F (physical; R² = 0.179), Q24A (environment; R² = 0.232), and Q3F (physical; R² = 0.258) show lower R² values, suggesting these items capture additional unique variance beyond their latent construct.

Parameter estimates

Factor loadings

Factor	Indicator	Symbol	Estimate	Std. Error	z-value	p	95% Confidence Interval	
							Lower	Upper
physical	Q3_F	λ11	0.508	0.026	19.363	< .001	0.456	0.559
	Q4_F	λ12	0.423	0.029	14.698	< .001	0.367	0.480
	Q10_F	λ13	0.817	0.012	66.850	< .001	0.793	0.841
	Q15_F	λ14	0.639	0.025	25.441	< .001	0.590	0.689
	Q16_F	λ15	0.593	0.023	25.768	< .001	0.548	0.638
psycho	Q17_F	λ16	0.921	0.007	123.217	< .001	0.907	0.936
	Q18_F	λ17	0.886	0.009	102.178	< .001	0.869	0.903
	Q5_P	λ21	0.740	0.017	44.811	< .001	0.708	0.773
	Q6_P	λ22	0.670	0.018	36.744	< .001	0.635	0.706
	Q7_P	λ23	0.664	0.018	36.636	< .001	0.629	0.700
social	Q11_P	λ24	0.659	0.020	33.747	< .001	0.621	0.697
	Q19_P	λ25	0.860	0.012	71.456	< .001	0.837	0.884
	Q26_P	λ26	0.640	0.020	32.579	< .001	0.602	0.679
	Q20_S	λ31	0.818	0.018	44.690	< .001	0.782	0.853
	Q21_S	λ32	0.583	0.027	21.430	< .001	0.530	0.636
environment	Q22_S	λ33	0.751	0.020	38.204	< .001	0.712	0.789
	Q8_A	λ41	0.510	0.028	18.083	< .001	0.455	0.566
	Q9_A	λ42	0.624	0.024	25.625	< .001	0.577	0.672
	Q12_A	λ43	0.712	0.021	34.427	< .001	0.672	0.753
	Q13_A	λ44	0.649	0.023	28.015	< .001	0.604	0.695
	Q14_A	λ45	0.766	0.020	38.821	< .001	0.727	0.804
	Q23_A	λ46	0.571	0.027	21.322	< .001	0.518	0.623
	Q24_A	λ47	0.482	0.028	17.030	< .001	0.426	0.537
	Q25_A	λ48	0.529	0.028	19.203	< .001	0.475	0.583

All standardized factor loadings are statistically significant ($p < .001$) and range from 0.423 (Q4F) to 0.921 (Q17F). Most loadings exceed 0.70, the threshold for acceptable indicator reliability, with 15 out of 24 items meeting this criterion. The physical domain shows the strongest indicators (Q17F = 0.921, Q18F = 0.886, Q10F = 0.817), while weaker loadings appear for Q4F (0.423), Q16F (0.593 physical), Q21S (0.583 social), and Q24A (0.482 environment). Items with loadings below 0.70 contribute less shared variance with their factors but may still provide valuable unique content.

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
physical	1.000	0.000			1.000	1.000
psycho	1.000	0.000			1.000	1.000
social	1.000	0.000			1.000	1.000
environment	1.000	0.000			1.000	1.000

Factor Covariances

		Estimate	Std. Error	z-value	p	95% Confidence Interval	
						Lower	Upper
physical	↔ psycho	0.909	0.010	90.825	< .001	0.890	0.929
physical	↔ social	0.638	0.024	26.546	< .001	0.591	0.685
physical	↔ environment	0.630	0.022	28.693	< .001	0.587	0.673
psycho	↔ social	0.771	0.019	41.000	< .001	0.734	0.808
psycho	↔ environment	0.725	0.021	34.939	< .001	0.684	0.766
social	↔ environment	0.550	0.028	19.549	< .001	0.495	0.605

The latent factor correlations reveal substantial relationships among quality of life domains. The physical-psycho correlation ($r = 0.909$) is particularly high, approaching multicollinearity concerns and suggesting potential empirical overlap between these constructs despite their theoretical distinctiveness. Moderate-to-strong correlations exist between psycho-social ($r = 0.771$), psycho-environment ($r = 0.725$), and physical-social ($r = 0.638$) domains. The weakest correlation is between social and environment ($r = 0.550$). These high intercorrelations (especially > 0.85) provide empirical justification for testing alternative models such as second-order or bifactor structures, which may better represent a general quality of life factor underlying these specific domains.

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Q3_F	0.742	0.000			0.742	0.742
Q4_F	0.821	0.000			0.821	0.821
Q10_F	0.333	0.000			0.333	0.333
Q15_F	0.591	0.000			0.591	0.591
Q16_F	0.649	0.000			0.649	0.649
Q17_F	0.151	0.000			0.151	0.151
Q18_F	0.214	0.000			0.214	0.214
Q5_P	0.452	0.000			0.452	0.452
Q6_P	0.551	0.000			0.551	0.551
Q7_P	0.559	0.000			0.559	0.559
Q11_P	0.566	0.000			0.566	0.566
Q19_P	0.260	0.000			0.260	0.260
Q26_P	0.590	0.000			0.590	0.590
Q20_S	0.332	0.000			0.332	0.332
Q21_S	0.660	0.000			0.660	0.660
Q22_S	0.436	0.000			0.436	0.436
Q8_A	0.740	0.000			0.740	0.740
Q9_A	0.610	0.000			0.610	0.610
Q12_A	0.492	0.000			0.492	0.492
Q13_A	0.578	0.000			0.578	0.578
Q14_A	0.414	0.000			0.414	0.414
Q23_A	0.675	0.000			0.675	0.675
Q24_A	0.768	0.000			0.768	0.768
Q25_A	0.720	0.000			0.720	0.720

Residual covariances

					95% Confidence Interval		
		Estimate	Std. Error	z-value	p	Lower	Upper
Q3_F	↔ Q4_F	0.388	0.025	15.711	< .001	0.339	0.436

Modification Indices

Cross-loadings

			Mod. Ind.	EPC
environment	→	Q5_P	175.788	0.503
physical	→	Q5_P	119.025	−1.039
environment	→	Q15_F	89.086	0.347
psycho	→	Q24_A	80.033	−0.333
physical	→	Q24_A	78.731	−0.289
psycho	→	Q10_F	55.182	0.631
environment	→	Q17_F	48.628	−0.251
environment	→	Q18_F	44.622	−0.236
environment	→	Q19_P	37.766	−0.254
social	→	Q24_A	33.765	−0.206
psycho	→	Q17_F	33.378	−0.549
psycho	→	Q14_A	31.896	0.221
physical	→	Q21_S	30.562	0.233
environment	→	Q10_F	25.193	0.167
physical	→	Q14_A	24.760	0.168
physical	→	Q7_P	23.918	0.438
physical	→	Q6_P	23.507	−0.442
physical	→	Q19_P	22.467	0.407
psycho	→	Q21_S	19.379	0.246
social	→	Q7_P	16.820	−0.237
social	→	Q10_F	15.951	0.169
physical	→	Q22_S	15.159	−0.224
social	→	Q14_A	14.544	0.139
psycho	→	Q18_F	14.423	−0.342
psycho	→	Q25_A	13.790	−0.139
social	→	Q16_F	13.052	0.159
psycho	→	Q16_F	11.037	0.311
social	→	Q6_P	10.854	0.172
physical	→	Q26_P	10.695	0.317
environment	→	Q26_P	10.545	−0.139

Modification indices (MI) identify potential model misspecifications. The largest MI values suggest cross-loadings: Q5P (psycho item) shows substantial cross-loading on environment (MI = 175.788, EPC = 0.503) and physical (MI = 119.025, EPC = 1.039), indicating this item shares variance with multiple factors. Additional cross-loadings include Q15F (physical) on environment (MI = 89.086), Q24A (environment) on psycho/physical (MI > 78), and Q10F (physical) on psycho (MI = 55.182). These suggest items may tap content from multiple quality of life domains, which is theoretically plausible given the interconnected nature of wellbeing facets. However, introducing cross-loadings would compromise model parsimony and theoretical clarity, particularly for CFA.

Residual covariances

			Mod. Ind.	EPC
Q5_P	↔	Q14_A	177.218	0.278
Q24_A	↔	Q25_A	124.552	0.282
Q17_F	↔	Q18_F	92.645	0.162
Q17_F	↔	Q5_P	44.099	−0.193
Q18_F	↔	Q5_P	41.613	−0.188
Q10_F	↔	Q18_F	34.964	−0.123
Q3_F	↔	Q15_F	32.487	0.181
Q23_A	↔	Q25_A	28.637	0.152
Q5_P	↔	Q6_P	26.979	0.113
Q15_F	↔	Q13_A	23.693	0.153
Q4_F	↔	Q15_F	21.772	0.148
Q10_F	↔	Q11_P	21.725	0.110
Q22_S	↔	Q23_A	19.739	0.133
Q23_A	↔	Q24_A	18.936	0.128
Q15_F	↔	Q25_A	18.429	0.127
Q14_A	↔	Q25_A	18.158	−0.138
Q10_F	↔	Q26_P	18.010	0.111
Q18_F	↔	Q19_P	17.850	0.077
Q15_F	↔	Q12_A	17.761	0.136
Q5_P	↔	Q12_A	17.225	0.104
Q15_F	↔	Q14_A	17.008	0.130
Q8_A	↔	Q9_A	15.076	0.108
Q16_F	↔	Q21_S	14.690	0.117
Q10_F	↔	Q17_F	13.924	−0.073
Q7_P	↔	Q24_A	13.335	−0.126
Q9_A	↔	Q14_A	13.266	−0.105
Q18_F	↔	Q14_A	13.001	−0.105
Q20_S	↔	Q22_S	12.810	0.124
Q5_P	↔	Q19_P	12.544	−0.093
Q12_A	↔	Q13_A	11.980	0.091
Q10_F	↔	Q7_P	11.865	0.078
Q15_F	↔	Q17_F	11.454	−0.104
Q18_F	↔	Q7_P	11.262	0.078
Q19_P	↔	Q24_A	11.261	−0.115
Q3_F	↔	Q6_P	11.084	−0.110
Q17_F	↔	Q19_P	10.935	0.059

Residual covariances indicate local dependencies between item pairs beyond what the common factor explains. The highest MI values occur for Q5P-Q14A (MI = 177.218, EPC = 0.278), Q24A-Q25A (MI = 124.552, EPC = 0.282), and Q17F-Q18F (MI = 92.645, EPC = 0.162). The Q17F-Q18F correlation is expected since both are physical domain items with similar content (pain/medical treatment). The Q3F-Q4F residual covariance was already freely estimated (see Residual covariances table), acknowledging their unique relationship. While freeing these residual correlations would improve fit, doing so should be theoretically justified (e.g., methodological artifacts, similar wording) rather than purely empirically driven, to avoid capitalization on chance.

Plots

Misfit plot

physical Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Greatest Lower Bound
Point estimate	0.819	0.876
95% CI lower bound	0.803	0.864
95% CI upper bound	0.836	0.891

psycho Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Greatest Lower Bound
Point estimate	0.823	0.840
95% CI lower bound	0.806	0.827
95% CI upper bound	0.839	0.861

social Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Greatest Lower Bound
Point estimate	0.696	0.696
95% CI lower bound	0.665	0.659
95% CI upper bound	0.727	0.729

environment Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Greatest Lower Bound
Point estimate	0.789	0.834
95% CI lower bound	0.769	0.821
95% CI upper bound	0.808	0.856

Reliability estimates assess internal consistency for each latent factor. McDonald's omega (ω) is preferred over Cronbach's alpha when using factor loadings from CFA, as it accounts for differential item weighting. The physical domain shows strong reliability ($\omega = 0.876$, 95% CI [0.864, 0.891]), as do psycho ($\omega = 0.840$) and environment ($\omega = 0.834$). The social domain exhibits lower but acceptable reliability ($\omega = 0.696$, 95% CI [0.659, 0.729]), likely due to having only three items and lower factor loadings. All reliabilities exceed the minimum threshold of 0.70, supporting adequate internal consistency. The Greatest Lower Bound (GLB) provides an upper-bound estimate, consistently higher than McDonald's omega, confirming that ω represents a conservative estimate of true reliability.

Model fit

Chi-square test

Model	X²	df	p
Baseline model	64072.987	253	
Factor model	862.260	207	< .001

The chi-square test ($\chi^2 = 862.260$, $df = 207$, $p < .001$) indicates significant discrepancy between the observed and model-implied covariance matrices. As with the four-factor model, chi-square remains highly sensitive to sample size ($n = 1047$), commonly rejecting well-fitting models in large samples. This test should not be used as the sole criterion for model evaluation. The baseline model (independence model) serves as a reference point for incremental fit indices.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.990
Tucker-Lewis Index (TLI)	0.987
Bentler-Bonett Non-normed Fit Index (NNFI)	0.987
Bentler-Bonett Normed Fit Index (NFI)	0.987
Parsimony Normed Fit Index (PNFI)	0.807
Bollen's Relative Fit Index (RFI)	0.984
Bollen's Incremental Fit Index (IFI)	0.990
Relative Noncentrality Index (RNI)	0.990

Information criteria

 Information criteria are only available with ML-type estimators

Akaike (AIC)

Bayesian (BIC)

Sample-size adjusted Bayesian (SSABIC)

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.055
RMSEA 90% CI lower bound	0.051
RMSEA 90% CI upper bound	0.059
RMSEA p-value	0.015
Standardized root mean square residual (SRMR)	0.049
Hoelter's critical N ($\alpha = .05$)	294.041
Hoelter's critical N ($\alpha = .01$)	313.071
Goodness of fit index (GFI)	0.991
McDonald fit index (MFI)	0.731
Expected cross validation index (ECVI)	

The bifactor model demonstrates superior fit compared to the four-factor correlated model. The CFI (0.990) and TLI (0.987) are higher than the four-factor model (CFI = 0.985, TLI = 0.983), and the RMSEA (0.055, 90% CI [0.051, 0.059]) is notably lower (four-factor RMSEA = 0.065). The SRMR (0.049) also shows improvement over the four-factor model (SRMR = 0.057). However, **traditional cutoff values must be interpreted with extreme caution**, as emphasized in Appendix Table 3 (Step 4). Bifactor models are structurally more complex and prone to overfitting, which can artificially inflate fit indices. The improved fit statistics do not automatically validate the bifactor structure - **theoretical justification and practical utility** must guide model selection. Fit indices should be interpreted as **effect size measures of misfit degree** rather than binary accept/reject criteria. Dynamic Fit Index (DFI) cutoffs tailored to this specific bifactor structure (accounting for ordinal data, DWLS estimation, and model complexity) provide more appropriate benchmarks than traditional cutoffs (see supplementary materials for DFI calculations).

	R²
Q3_F	0.655
Q4_F	0.489
Q10_F	0.630
Q15_F	0.425
Q16_F	0.332
Q17_F	0.838
Q18_F	0.776
Q6_P	0.884
Q7_P	0.449
Q11_P	0.443
Q19_P	0.732
Q26_P	0.420
Q20_S	0.678
Q21_S	0.276
Q22_S	0.614
Q8_A	0.228
Q9_A	0.355
Q12_A	0.485
Q13_A	0.418
Q14_A	0.472
Q23_A	0.364
Q24_A	0.424
Q25_A	0.375

The R² values in the bifactor model differ substantially from the four-factor correlated model due to the dual-loading structure. Each item's variance is partitioned between a general quality of life factor (qol) and its specific domain factor (physical, psycho, social, environment). Items Q17F (R² = 0.865), Q18F (R² = 0.811), Q10F (R² = 0.678), and Q19P (R² = 0.804) show the highest total explained variance. Notably, items Q3F (R² = 0.577), Q4F (R² = 0.471), Q6P (R² = 0.469), and Q20S (R² = 0.726) demonstrate substantial increases in R² compared to the four-factor model, suggesting these items capture considerable general quality of life variance beyond their specific domains. Items with high R² values in bifactor models indicate strong relationships with the general factor, while items with lower R² may retain more unique domain-specific variance. The bifactor structure provides insights into whether items primarily reflect general quality of life or domain-specific facets.

Parameter estimates

Factor loadings

Factor	Indicator	Symbol	Estimate	Std. Error	z-value	p	95% Confidence Interval	
							Lower	Upper
physical	Q3_F	λ11	0.690	0.042	16.345	< .001	0.607	0.773
	Q4_F	λ12	0.606	0.039	15.419	< .001	0.529	0.684
	Q10_F	λ13	0.116	0.030	3.844	< .001	0.057	0.175
	Q15_F	λ14	0.290	0.037	7.942	< .001	0.219	0.362
	Q16_F	λ15	0.059	0.037	1.611	0.107	−0.013	0.131
psycho	Q17_F	λ16	0.268	0.031	8.770	< .001	0.208	0.328
	Q18_F	λ17	0.257	0.031	8.338	< .001	0.196	0.317
	Q6_P	λ21	0.696	0.479	1.454	0.146	−0.242	1.635
	Q7_P	λ22	0.059	0.051	1.158	0.247	−0.041	0.159
	Q11_P	λ23	0.021	0.032	0.672	0.502	−0.041	0.084
social	Q19_P	λ24	0.085	0.066	1.282	0.200	−0.045	0.214
	Q26_P	λ25	0.135	0.102	1.330	0.184	−0.064	0.335
	Q20_S	λ31	0.566	0.052	10.952	< .001	0.465	0.667
	Q21_S	λ32	0.269	0.035	7.611	< .001	0.199	0.338
	Q22_S	λ33	0.567	0.052	10.845	< .001	0.464	0.669
environment	Q8_A	λ41	0.300	0.032	9.446	< .001	0.238	0.362
	Q9_A	λ42	0.386	0.029	13.488	< .001	0.330	0.442
	Q12_A	λ43	0.485	0.026	18.608	< .001	0.434	0.536
	Q13_A	λ44	0.453	0.028	16.478	< .001	0.399	0.507
	Q14_A	λ45	0.441	0.028	15.757	< .001	0.386	0.496
qol	Q23_A	λ46	0.468	0.029	16.353	< .001	0.412	0.524
	Q24_A	λ47	0.603	0.028	21.426	< .001	0.548	0.658
	Q25_A	λ48	0.518	0.027	18.868	< .001	0.464	0.571
	Q3_F	λ51	0.422	0.030	13.966	< .001	0.363	0.481
	Q4_F	λ52	0.349	0.031	11.186	< .001	0.288	0.410
	Q10_F	λ53	0.785	0.014	57.090	< .001	0.758	0.812
	Q15_F	λ54	0.584	0.027	21.563	< .001	0.531	0.637
	Q16_F	λ55	0.573	0.023	24.897	< .001	0.528	0.618
	Q17_F	λ56	0.875	0.012	74.814	< .001	0.852	0.898
	Q18_F	λ57	0.842	0.013	66.801	< .001	0.818	0.867
	Q6_P	λ58	0.632	0.022	29.265	< .001	0.589	0.674
	Q7_P	λ59	0.668	0.019	35.513	< .001	0.631	0.704
	Q11_P	λ510	0.666	0.020	33.248	< .001	0.626	0.705
	Q19_P	λ511	0.851	0.012	71.086	< .001	0.828	0.875
	Q26_P	λ512	0.634	0.022	29.242	< .001	0.591	0.676
	Q20_S	λ513	0.598	0.021	28.122	< .001	0.556	0.640
	Q21_S	λ514	0.452	0.026	17.521	< .001	0.401	0.502
	Q22_S	λ515	0.541	0.024	22.910	< .001	0.495	0.588
	Q8_A	λ516	0.372	0.028	13.122	< .001	0.316	0.427
	Q9_A	λ517	0.454	0.026	17.366	< .001	0.403	0.505
	Q12_A	λ518	0.499	0.024	20.475	< .001	0.451	0.547
	Q13_A	λ519	0.461	0.025	18.203	< .001	0.411	0.511
	Q14_A	λ520	0.527	0.024	21.667	< .001	0.479	0.574
	Q23_A	λ521	0.380	0.029	13.230	< .001	0.324	0.436
	Q24_A	λ522	0.246	0.031	7.807	< .001	0.184	0.307
	Q25_A	λ523	0.328	0.029	11.254	< .001	0.271	0.385

The bifactor model presents a complex dual-loading structure where each item loads simultaneously on the general quality of life factor (qol) and its specific domain factor. **General factor loadings** are consistently high and significant across most items, ranging from 0.246 (Q24A) to 0.875 (Q17F), with 19 out of 23 items exceeding 0.40. This suggests a strong general quality of life dimension underlies all items. **Specific factor loadings** vary considerably: physical domain shows moderate specific loadings (Q3F = 0.690, Q4F = 0.606) but very weak loadings for Q10F (0.116) and Q16F (0.059, $p = .107$, non-significant), indicating these items primarily measure general quality of life rather than specific physical aspects. The psycho domain exhibits peculiar results with non-significant specific loadings for most items (Q6P, Q7P, Q11P, Q19P, Q26P all $p > .05$), suggesting the psychological domain may be almost entirely subsumed by the general factor. Social and environment domains retain moderate specific loadings beyond the general factor. **Non-significant specific loadings** (e.g., Q16F, all psycho items) indicate these items provide little domain-specific information after controlling for the general factor, questioning whether these specific factors are empirically distinguishable from the general factor in this sample.

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
physical	1.000	0.000			1.000	1.000
psycho	1.000	0.000			1.000	1.000
social	1.000	0.000			1.000	1.000
environment	1.000	0.000			1.000	1.000
qol	1.000	0.000			1.000	1.000

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Q3_F	0.345	0.000			0.345	0.345
Q4_F	0.511	0.000			0.511	0.511
Q10_F	0.370	0.000			0.370	0.370
Q15_F	0.575	0.000			0.575	0.575
Q16_F	0.668	0.000			0.668	0.668
Q17_F	0.162	0.000			0.162	0.162
Q18_F	0.224	0.000			0.224	0.224
Q6_P	0.116	0.000			0.116	0.116
Q7_P	0.551	0.000			0.551	0.551
Q11_P	0.557	0.000			0.557	0.557
Q19_P	0.268	0.000			0.268	0.268
Q26_P	0.580	0.000			0.580	0.580
Q20_S	0.322	0.000			0.322	0.322
Q21_S	0.724	0.000			0.724	0.724
Q22_S	0.386	0.000			0.386	0.386
Q8_A	0.772	0.000			0.772	0.772
Q9_A	0.645	0.000			0.645	0.645
Q12_A	0.515	0.000			0.515	0.515
Q13_A	0.582	0.000			0.582	0.582
Q14_A	0.528	0.000			0.528	0.528
Q23_A	0.636	0.000			0.636	0.636
Q24_A	0.576	0.000			0.576	0.576
Q25_A	0.625	0.000			0.625	0.625

Modification Indices

Cross-loadings

			Mod. Ind.	EPC
environment	→	Q15_F	114.813	0.300
environment	→	Q18_F	46.170	-0.177
environment	→	Q22_S	45.651	0.170
environment	→	Q17_F	42.693	-0.170
social	→	Q23_A	36.821	0.232
social	→	Q6_P	31.214	0.184
physical	→	Q20_S	23.444	-0.174
psycho	→	Q20_S	19.965	0.164
social	→	Q24_A	18.501	0.172
physical	→	Q6_P	17.324	-0.154
environment	→	Q3_F	15.960	0.119
physical	→	Q22_S	12.516	-0.128
social	→	Q19_P	11.241	0.108
social	→	Q17_F	11.069	-0.115
environment	→	Q4_F	10.494	0.093
physical	→	Q11_P	10.290	0.111

Modification indices (MI) for the bifactor model are notably lower than those in the four-factor correlated model, reflecting improved model fit. The highest MI values suggest cross-loadings: Q15F (physical item) on environment (MI = 114.813, EPC = 0.300), Q18F and Q22S on environment (MI > 40), and Q17F on environment (MI = 42.693). These cross-loadings indicate that some physical items share variance with environmental aspects beyond what the general and specific factors capture. Notably, item Q5 was excluded from this analysis due to its problematic behavior in the four-factor model (extensive cross-loadings and high MI values). The lower MI values overall suggest that the bifactor structure better accommodates the empirical covariance patterns, though this does not necessarily validate its theoretical superiority. As with the four-factor model, introducing cross-loadings would compromise parsimony and interpretability, particularly problematic in bifactor models where complexity is already high.

Residual covariances

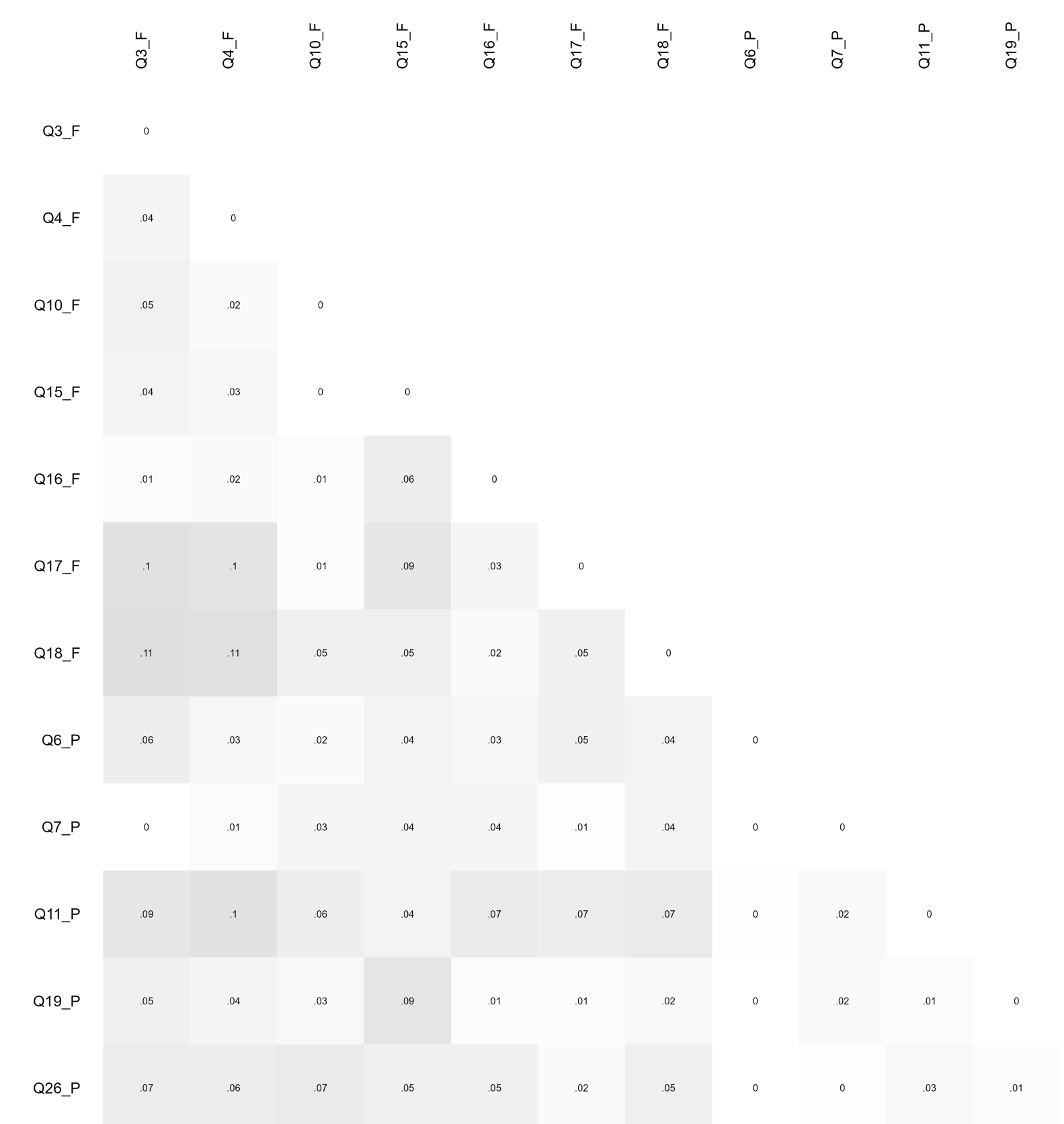
			Mod. Ind.	EPC
Q17_F	↔	Q18_F	155.946	0.215
Q3_F	↔	Q4_F	58.434	0.819
Q24_A	↔	Q25_A	44.606	0.231
Q12_A	↔	Q14_A	31.078	0.152
Q22_S	↔	Q23_A	30.368	0.165
Q8_A	↔	Q9_A	29.681	0.156
Q3_F	↔	Q17_F	29.501	-0.244
Q3_F	↔	Q18_F	28.621	-0.239
Q15_F	↔	Q25_A	26.451	0.154
Q4_F	↔	Q18_F	23.224	-0.198
Q4_F	↔	Q17_F	22.166	-0.192
Q15_F	↔	Q13_A	21.862	0.149
Q12_A	↔	Q13_A	20.411	0.132
Q14_A	↔	Q25_A	19.716	-0.155
Q18_F	↔	Q14_A	18.537	-0.128
Q15_F	↔	Q14_A	17.741	0.134
Q15_F	↔	Q12_A	16.994	0.135
Q6_P	↔	Q20_S	16.792	0.103
Q13_A	↔	Q14_A	14.741	0.109
Q10_F	↔	Q18_F	13.291	-0.076
Q22_S	↔	Q24_A	12.672	0.113
Q12_A	↔	Q25_A	12.573	-0.121
Q4_F	↔	Q11_P	12.127	0.112
Q15_F	↔	Q17_F	11.931	-0.109
Q10_F	↔	Q26_P	11.389	0.091
Q14_A	↔	Q24_A	10.792	-0.115
Q9_A	↔	Q23_A	10.654	0.101
Q17_F	↔	Q12_A	10.534	-0.097
Q18_F	↔	Q11_P	10.493	-0.087
Q17_F	↔	Q22_S	10.155	-0.092
Q18_F	↔	Q12_A	10.094	-0.096
Q4_F	↔	Q12_A	10.033	0.098
Q18_F	↔	Q13_A	10.003	-0.095

Residual covariances in the bifactor model show fewer and smaller local dependencies compared to the four-factor correlated model. The highest MI values occur for Q17F-Q18F (MI = 155.946, EPC = 0.215), Q3F-Q4F (MI = 58.434), Q24A-Q25A (MI = 44.606), and various other item pairs. The Q17F-Q18F pair (both physical domain items about pain and medical treatment) consistently shows residual correlation across all models, suggesting shared method variance or semantic similarity beyond the latent structure. The Q3F-Q4F residual covariance was already freely estimated in the model specification, acknowledging these two reverse-coded items share unique variance. Overall, the bifactor structure reduces residual dependencies, as the general factor absorbs much of the covariance among items. However, persistent residual correlations suggest that even the bifactor model does not fully capture all item relationships, indicating potential item-specific nuances or minor method effects.

Plots

Important limitation: JASP version 0.17.1 does not compute reliability coefficients for bifactor models directly in the output. Unlike the four-factor correlated model where McDonald's omega and GLB were calculated for each domain, bifactor models require distinct reliability metrics: **omega-hierarchical (ω_H)** for the general factor, **omega-subscale (ω_S)** for specific factors, and **explained common variance (ECV)** to assess the proportion of common variance attributable to the general versus specific factors. These metrics are crucial for evaluating whether the bifactor structure is empirically justified - high ω_H and ECV values indicate a strong general factor, while substantial ω_S values suggest specific factors retain meaningful reliable variance. Without these metrics, interpretation of the bifactor model remains incomplete. **Researchers must use lavaan or specialized R packages** (e.g., psych, BifactorIndicesCalculator, semTools) to compute these indices. Additionally, researchers should examine **omega-hierarchical for subscales (ω_{HS})** to determine if domain scores (e.g., physical, psychological) provide reliable information beyond the general factor. If ω_{HS} values are low ($< .30$), specific domain scores may be unreliable for interpretation, suggesting that only the general quality of life score should be used in practice.

Misfit plot



Q20_S	.06	.08	.06	.07	0	.03	.03	.09	.01	.02	.06
Q21_S	.04	.01	.01	.02	.07	.01	0	.03	.05	0	.02
Q22_S	.02	.04	.06	.05	.01	.08	.05	.07	.06	.01	.03
Q8_A	.04	.02	.02	.06	.04	.06	.05	.01	.07	.01	.02
Q9_A	.04	.05	.04	.07	.01	.08	.06	.01	.03	.02	.01
Q12_A	.05	.09	.03	.12	.01	.08	.08	.05	.02	.04	.03
Q13_A	.04	.04	.02	.14	.02	.06	.08	.02	.03	.03	.06
Q14_A	.04	.06	.04	.12	.03	.07	.11	.02	.05	0	.03
Q23_A	.02	.02	.01	.08	.02	.09	.07	.02	.04	0	0
Q24_A	.05	.01	.03	.07	0	.02	.02	.03	.05	.01	.02
Q25_A	.07	.02	.05	.14	.05	.03	.02	.01	0	.01	.01

Model fit

	AIC	BIC	n	Baseline test			Difference test		
				χ^2	df	p	$\Delta\chi^2$	Δ df	p
Model 1			1047	1007.178	226	< .001	1007.178	226	< .001

Note. The AIC, BIC and additional information criteria are only available with ML-type estimators

The significant chi-square indicates discrepancy between observed and model-implied covariance matrices. However, as consistently emphasized, chi-square tests are highly sensitive to sample size (n = 1047) and should not serve as the sole basis for model evaluation. The second-order model constrains the four first-order factors (physical, psycho, social, environment) to be explained by a higher-order general quality of life factor (QOL), representing a more parsimonious hierarchical structure compared to the freely correlated four-factor model.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.988
T-size CFI	0.986
Tucker-Lewis Index (TLI)	0.986
Bentler-Bonett Non-normed Fit Index (NNFI)	0.986
Bentler-Bonett Normed Fit Index (NFI)	0.984
Parsimony Normed Fit Index (PNFI)	0.879
Bollen's Relative Fit Index (RFI)	0.982
Bollen's Incremental Fit Index (IFI)	0.988
Relative Noncentrality Index (RNI)	0.988

Note. T-size CFI is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: Model 1

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.057
RMSEA 90% CI lower bound	0.054
RMSEA 90% CI upper bound	0.061
RMSEA p-value	3.411×10 ⁻⁴
T-size RMSEA	0.061
Standardized root mean square residual (SRMR)	0.054
Hoelter's critical N ($\alpha = .05$)	273.171
Hoelter's critical N ($\alpha = .01$)	290.109
Goodness of fit index (GFI)	0.989
McDonald fit index (MFI)	0.688
Expected cross validation index (ECVI)	

Note. T-size RMSEA is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: Model 1

The second-order model demonstrates fit indices comparable to the four-factor correlated model but slightly inferior to the bifactor model. The CFI (0.988) and TLI (0.986) are strong, the RMSEA (0.057, 90% CI [0.054, 0.061]) is acceptable, and the SRMR (0.054) suggests minimal discrepancy between observed and predicted correlations. Compared to the four-factor model (CFI = 0.985, RMSEA = 0.065), the second-order model shows marginal improvement, while the bifactor model exhibits superior fit (CFI = 0.990, RMSEA = 0.055). However, **traditional cutoff values must be interpreted with caution**, as discussed in Appendix Table 3 (Step 4). These conventional thresholds were derived from simulations using ML estimation with continuous data and may not generalize to DWLS estimation with ordinal data. Furthermore, second-order models introduce additional constraints (first-order factors are regressed on a higher-order factor), which affects degrees of freedom and fit index behavior. Fit indices should be interpreted as **effect size measures reflecting degree of misfit** rather than binary decision rules. Dynamic Fit Index (DFI) cutoffs tailored to this second-order structure provide more appropriate benchmarks (see supplementary materials). The T-size CFI (0.986) and T-size RMSEA (0.061) adjust for model complexity and sample size, offering alternative perspectives on model adequacy.

	R²
psycho	1.000
Q6_P	0.436
Q7_P	0.460
Q11_P	0.450
Q19_P	0.761
Q26_P	0.420
Q3_F	0.260
Q4_F	0.178
Q10_F	0.657
Q15_F	0.413
Q16_F	0.348
Q17_F	0.852
Q18_F	0.789
Q20_S	0.670
Q21_S	0.346
Q22_S	0.558
Q8_A	0.261
Q9_A	0.397
Q12_A	0.511
Q13_A	0.439
Q14_A	0.535
Q23_A	0.336
Q24_A	0.242
Q25_A	0.293
physical	0.832
social	0.548
environment	0.469

The R² values in the second-order model have dual interpretations. **For observed indicators**, R² represents variance explained by their respective first-order factors, showing values nearly identical to the four-factor correlated model (e.g., Q17F = 0.852, Q18F = 0.789, Q10F = 0.657). **For first-order latent factors**, R² indicates variance explained by the second-order QOL factor: physical (R² = 0.832) shows strong explanation by the higher-order factor; social (R² = 0.548) retains moderate unique variance beyond QOL; and environment (R² = 0.469) demonstrates substantial domain-specific variance not captured by the general factor. High R² values for first-order factors justify the hierarchical structure, while lower values (social, environment) suggest these domains provide unique information beyond general quality of life.

Parameter estimates

Factor Loadings

Latent	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
						Lower	Upper	All	LV	Endo
environment	Q8_A	1.000	0.000			1.000	1.000	0.510	0.510	0.510
	Q9_A	1.234	0.078	15.732	< .001	1.080	1.388	0.630	0.630	0.630
	Q12_A	1.401	0.088	15.882	< .001	1.228	1.574	0.715	0.715	0.715
	Q13_A	1.298	0.086	15.037	< .001	1.129	1.468	0.663	0.663	0.663
	Q14_A	1.433	0.090	15.929	< .001	1.256	1.609	0.731	0.731	0.731
	Q23_A	1.135	0.081	13.958	< .001	0.976	1.295	0.579	0.579	0.579
	Q24_A	0.963	0.075	12.769	< .001	0.815	1.111	0.492	0.492	0.492
	Q25_A	1.060	0.077	13.750	< .001	0.909	1.211	0.541	0.541	0.541
physical	Q3_F	1.000	0.000			1.000	1.000	0.509	0.509	0.509
	Q4_F	0.828	0.057	14.643	< .001	0.718	0.939	0.422	0.422	0.422
	Q10_F	1.591	0.080	19.823	< .001	1.434	1.749	0.811	0.811	0.811
	Q15_F	1.261	0.071	17.707	< .001	1.121	1.400	0.642	0.642	0.642
	Q16_F	1.157	0.070	16.418	< .001	1.019	1.295	0.590	0.590	0.590
	Q17_F	1.811	0.092	19.626	< .001	1.630	1.992	0.923	0.923	0.923
	Q18_F	1.743	0.089	19.580	< .001	1.568	1.917	0.888	0.888	0.888
	Q6_P	1.000	0.000			1.000	1.000	0.660	0.660	0.660
psycho	Q7_P	1.027	0.039	26.007	< .001	0.950	1.105	0.678	0.678	0.678
	Q11_P	1.017	0.039	26.076	< .001	0.941	1.093	0.671	0.671	0.671
	Q19_P	1.322	0.041	32.224	< .001	1.241	1.402	0.872	0.872	0.872
	Q26_P	0.982	0.038	25.960	< .001	0.908	1.056	0.648	0.648	0.648
	QOL	1.000	0.000			1.000	1.000	0.912	0.912	0.912
social	physical	1.420	0.086	16.526	< .001	1.252	1.588	1.000	1.000	1.000
	psycho	1.304	0.082	15.971	< .001	1.144	1.464	0.740	0.740	0.740
	social	0.752	0.062	12.074	< .001	0.630	0.874	0.685	0.685	0.685
	environment	1.000	0.000			1.000	1.000	0.818	0.818	0.818
	Q20_S	0.718	0.038	19.118	< .001	0.645	0.792	0.588	0.588	0.588
social	Q21_S	0.913	0.036	25.194	< .001	0.842	0.984	0.747	0.747	0.747
	Q22_S									

The second-order model presents a hierarchical structure where observed indicators load on first-order factors (physical, psycho, social, environment), which in turn load on the second-order QOL factor. **Indicator loadings on first-order factors** are nearly identical to the four-factor correlated model, ranging from 0.422 (Q4F) to 0.923 (Q17F), maintaining the same pattern of strong loadings for physical and psycho domains and weaker loadings for some social and environment items. **First-order factor loadings on the second-order QOL factor** reveal the hierarchical structure: psycho shows loading = 1.000 (standardized), physical = 0.912, social = 0.740, and environment = 0.685. The loading = 1.000 for psycho is **an artifact of the identification constraint** (psycho variance fixed at zero), not a substantive finding. In second-order models, first-order factors serve as indicators of the higher-order construct, and one factor must be constrained for identification - this is analogous to setting one indicator loading to 1.0 in first-order CFA. Physical demonstrates strong loading on QOL, while social and environment show moderate loadings, suggesting these domains retain more unique variance beyond the general construct. The second-order structure assumes first-order factors fully mediate the relationship between QOL and observed indicators, a more restrictive assumption than the bifactor model's dual-loading structure.

Factor variances

Variable	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
					Lower	Upper	All	LV	Endo
psycho	0.000	0.000			0.000	0.000	0.000	0.000	0.000
physical	0.044	0.006	7.418	< .001	0.032	0.055	0.168	0.168	0.168
social	0.303	0.025	12.191	< .001	0.254	0.351	0.452	0.452	0.452
environment	0.138	0.017	8.255	< .001	0.106	0.171	0.531	0.531	0.531
QOL	0.216	0.024	9.179	< .001	0.170	0.262	1.000	1.000	1.000

In the second-order model, factor variances reveal the scaling and identification strategy employed. The psycho factor variance is constrained to zero (0.000) for model identification purposes, as indicated in the model specification. This constraint is necessary because the loading of psycho on QOL was freely estimated. The second-order QOL factor variance is set to 1.000 for scaling purposes, following the marker variable approach. Other first-order factors (physical, social, environment) have estimated variances (physical = 0.168, social = 0.452, environment = 0.531), representing their unique variance not explained by the second-order QOL factor. These residual variances correspond to 1 - R² values from the R-Squared table: environment (0.531 ≈ 1 - 0.469), social (0.452 ≈ 1 - 0.548), and physical (0.168 ≈ 1 - 0.832).

Residual variances

Variable	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
					Lower	Upper	All	LV	Endo
Q6_P	0.564	0.000			0.564	0.564	0.564	0.564	0.564
Q7_P	0.540	0.000			0.540	0.540	0.540	0.540	0.540
Q11_P	0.550	0.000			0.550	0.550	0.550	0.550	0.550
Q19_P	0.239	0.000			0.239	0.239	0.239	0.239	0.239
Q26_P	0.580	0.000			0.580	0.580	0.580	0.580	0.580
Q3_F	0.740	0.000			0.740	0.740	0.740	0.740	0.740
Q4_F	0.822	0.000			0.822	0.822	0.822	0.822	0.822
Q10_F	0.343	0.000			0.343	0.343	0.343	0.343	0.343
Q15_F	0.587	0.000			0.587	0.587	0.587	0.587	0.587
Q16_F	0.652	0.000			0.652	0.652	0.652	0.652	0.652
Q17_F	0.148	0.000			0.148	0.148	0.148	0.148	0.148
Q18_F	0.211	0.000			0.211	0.211	0.211	0.211	0.211
Q20_S	0.330	0.000			0.330	0.330	0.330	0.330	0.330
Q21_S	0.654	0.000			0.654	0.654	0.654	0.654	0.654
Q22_S	0.442	0.000			0.442	0.442	0.442	0.442	0.442
Q8_A	0.739	0.000			0.739	0.739	0.739	0.739	0.739
Q9_A	0.603	0.000			0.603	0.603	0.603	0.603	0.603
Q12_A	0.489	0.000			0.489	0.489	0.489	0.489	0.489
Q13_A	0.561	0.000			0.561	0.561	0.561	0.561	0.561
Q14_A	0.465	0.000			0.465	0.465	0.465	0.465	0.465
Q23_A	0.664	0.000			0.664	0.664	0.664	0.664	0.664
Q24_A	0.758	0.000			0.758	0.758	0.758	0.758	0.758
Q25_A	0.707	0.000			0.707	0.707	0.707	0.707	0.707

Residual variances for observed indicators represent unique variance not explained by their first-order factors, mirroring values from the four-factor correlated model. These range from 0.148 (Q17F) to 0.822 (Q4F), corresponding to 1 - R² values. Items with low residual variances (Q17F = 0.148, Q18F = 0.211, Q10F = 0.343) are strongly explained by their factors, while items with high residual variances (Q4F = 0.822, Q3F = 0.740, Q24A = 0.758) contain substantial unique measurement error or specific variance. The consistency of residual variances across the four-factor and second-order models confirms that the hierarchical structure primarily affects relationships among first-order factors rather than indicator-factor relationships. However, this also indicates that the second-order model does not improve item-level measurement precision compared to the simpler four-factor structure.

Residual covariances

Variables	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
					Lower	Upper	All	LV	Endo
Q3_F - Q4_F	0.387	0.025	15.655	< .001	0.339	0.436	0.497	0.387	0.497

The second-order model includes one freely estimated residual covariance between Q3F and Q4F (estimate = 0.387, p < .001), identical to the four-factor correlated model. This acknowledges the unique relationship between these two reverse-coded items within the physical domain, likely reflecting method effects or semantic similarity beyond the latent structure. The significant residual covariance suggests these items share variance not captured by the physical factor or the second-order QOL factor. No other residual covariances were freely estimated in this model, maintaining parsimony. The modification indices table (if examined) would reveal potential additional residual correlations that could improve fit, though introducing these should be theoretically justified rather than purely data-driven.

Covariance tables

Standardized residuals matrix

Q6_P	Q7_P	Q11_P	Q19_P	Q26_P	Q3_F	Q4_F	Q10_F	Q15_F	Q16_F	Q17_F	Q18_F	Q20_S	Q21_S	Q22_S	Q8_A	Q9_A
< .001																
1.087	< .001															
< .001	0.581	< .001														
1.723	< .001	< .001	< .001													
3.178	< .001	< .001	< .001	< .001												
< .001	< .001	2.374	< .001	1.340	< .001											
< .001	< .001	3.139	< .001	1.412	< .001	< .001										
1.732	4.501	6.150	< .001	5.332	2.632	1.063	< .001									
< .001	< .001	1.773	< .001	< .001	6.506	5.308	< .001	< .001								
< .001	< .001	< .001	2.225	2.967	< .001	0.156	< .001	< .001	< .001							
< .001	0.471	< .001	2.923	< .001	< .001	< .001	< .001	< .001	0.701	< .001						
< .001	3.539	< .001	4.727	< .001	< .001	< .001	< .001	< .001	< .001	8.346	< .001					
4.143	< .001	< .001	2.410	< .001	< .001	< .001	< .001	< .001	0.643	< .001	< .001	< .001				
1.362	< .001	0.652	1.835	0.387	1.042	< .001	2.491	< .001	3.966	1.165	1.943	< .001	< .001			
2.914	< .001	< .001	0.397	< .001	< .001	< .001	< .001	< .001	0.014	< .001	< .001	3.746	< .001	< .001	< .001	
0.904	4.001	1.093	< .001	0.619	1.119	0.520	2.754	2.567	< .001	< .001	< .001	0.975	< .001	< .001	< .001	< .001
0.360	1.873	1.427	0.282	0.129	1.072	1.575	4.188	3.126	0.875	< .001	< .001	< .001	< .001	1.382	4.044	< .001
< .001	< .001	2.236	< .001	0.797	1.410	3.272	3.517	5.247	0.554	< .001	< .001	< .001	0.794	2.396	< .001	< .001
0.350	1.628	1.617	< .001	< .001	0.990	1.024	0.604	5.998	< .001	< .001	< .001	1.796	< .001	2.078	< .001	< .001
1.296	< .001	0.559	< .001	< .001	1.358	2.004	5.139	5.860	2.797	< .001	< .001	0.829	0.163	3.430	< .001	< .001
0.037	< .001	< .001	< .001	< .001	< .001	< .001	< .001	2.306	< .001	< .001	< .001	3.020	0.152	5.745	< .001	2.000
< .001	< .001	< .001	< .001	< .001	< .001	< .001	< .001	0.709	< .001	< .001	< .001	0.040	0.245	2.083	0.294	< .001
< .001	< .001	< .001	< .001	< .001	1.237	< .001	< .001	4.974	1.274	< .001	< .001	0.392	0.662	0.413	0.526	< .001

Interpretation and model identification considerations: The second-order model offers a parsimonious hierarchical representation where general quality of life (QOL) explains correlations among four specific domains. This structure is theoretically appealing when domains are conceptualized as manifestations of an overarching construct. In this specification, the psycho factor serves as the reference indicator for the second-order factor (variance constrained to zero for identification), a standard and necessary practice in hierarchical models. The choice to constrain psycho rather than another factor is arbitrary from a statistical perspective; alternative identification strategies (e.g., constraining physical, social, or environment) would yield equivalent model fit but different parameter estimates. Substantively, the model reveals that physical quality of life is strongly determined by general QOL ($R^2 = 0.832$), while social ($R^2 = 0.548$) and environment ($R^2 = 0.469$) retain considerable domain-specific variance, suggesting these facets provide unique information beyond the general construct. Researchers should compare this model against alternatives (four-factor correlated, bifactor) using not only fit indices but also theoretical interpretability, parameter stability, and practical utility. The second-order model is most appropriate when domains are conceptualized as intermediate variables fully mediating the effect of a general construct, and when psychometric evidence supports this hierarchical structure across multiple samples.

psycho Reliability without Q5_F

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Greatest Lower Bound
Point estimate	0.796	0.809
95% CI lower bound	0.777	0.793
95% CI upper bound	0.816	0.831

This analysis examines the internal consistency of the psychological domain after excluding item Q5_F ("How much do you enjoy life?"), which exhibited problematic cross-loadings in the four-factor model (MI > 175 with environment and physical factors). The reliability remains strong: McDonald's omega (ω = 0.796, 95% CI [0.777, 0.816]) and Greatest Lower Bound (GLB = 0.809, 95% CI [0.793, 0.831]) both exceed the acceptable threshold of 0.70, indicating adequate internal consistency for the psychological domain with the remaining five items (Q6P, Q7P, Q11P, Q19P, Q26P). Notably, omega decreased slightly from the original six-item version (ω = 0.823 in the CFA_4F model) to this five-item version (ω = 0.796), a reduction of 0.027 points. This minimal decrease suggests that Q5_F contributed to reliability despite its cross-loading issues, though its removal may improve construct validity by reducing multidimensionality. The GLB consistently exceeds omega, providing an upper-bound estimate and confirming omega represents a conservative reliability estimate. Researchers should evaluate whether the trade-off between slightly lower reliability and improved discriminant validity (by removing Q5_F) is justified based on theoretical considerations and the importance of distinguishing psychological quality of life from other domains in their specific application.

Model fit

Chi-square test

Model	X²	df	p
Baseline model	64072.987	253	
Factor model	977.820	223	< .001

The chi-square test ($\chi^2 = 977.820$, $df = 223$, $p < .001$) indicates significant discrepancy between the observed and model-implied covariance matrices for the four-factor model excluding item Q5P. Compared to the original 4F model ($\chi^2 = 1339.798$, $df = 245$), this model shows improved chi-square value despite having fewer degrees of freedom, reflecting better fit after removing the problematic item. However, chi-square remains highly sensitive to sample size ($n = 1047$) and should not be the primary criterion for model evaluation. The baseline model serves as the independence model reference point.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.988
Tucker-Lewis Index (TLI)	0.987
Bentler-Bonett Non-normed Fit Index (NNFI)	0.987
Bentler-Bonett Normed Fit Index (NFI)	0.985
Parsimony Normed Fit Index (PNFI)	0.868
Bollen's Relative Fit Index (RFI)	0.983
Bollen's Incremental Fit Index (IFI)	0.988
Relative Noncentrality Index (RNI)	0.988

Information criteria

 Information criteria are only available with ML-type estimators

Akaike (AIC)

Bayesian (BIC)

Sample-size adjusted Bayesian (SSABIC)

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.057
RMSEA 90% CI lower bound	0.053
RMSEA 90% CI upper bound	0.061
RMSEA p-value	9.592×10 ⁻⁴
Standardized root mean square residual (SRMR)	0.053
Hoelter's critical N ($\alpha = .05$)	277.884
Hoelter's critical N ($\alpha = .01$)	295.227
Goodness of fit index (GFI)	0.989
McDonald fit index (MFI)	0.697
Expected cross validation index (ECVI)	

The four-factor model without Q5P demonstrates fit indices nearly identical to the model including Q5P, with marginal improvements. The CFI (0.988) slightly exceeds the original model (CFI = 0.985), the TLI (0.987) also improves from 0.983, and the RMSEA (0.057, 90% CI [0.053, 0.061]) decreases from 0.065. The SRMR (0.053) improves modestly from 0.057. These improvements suggest that removing Q5P, which exhibited extensive cross-loadings ($MI > 175$ with environment and physical), enhanced model parsimony without sacrificing fit. However, **traditional cutoff values must be interpreted with caution**, as emphasized in Appendix Table 3 (Step 4). These conventional thresholds (RMSEA < .06, CFI > .95) were derived from ML estimation with continuous data and may not generalize to DWLS estimation with ordinal variables. Fit indices should be interpreted as **effect size measures reflecting degree of misfit** rather than binary accept/reject criteria. Dynamic Fit Index (DFI) cutoffs tailored to this 23-item, four-factor ordinal structure provide more appropriate benchmarks (see supplementary materials). The fit improvement is modest, indicating Q5P removal primarily benefits discriminant validity rather than overall model-data correspondence.

	R²
Q3_F	0.260
Q4_F	0.179
Q10_F	0.657
Q15_F	0.414
Q16_F	0.348
Q17_F	0.851
Q18_F	0.788
Q6_P	0.435
Q7_P	0.459
Q11_P	0.450
Q19_P	0.755
Q26_P	0.420
Q20_S	0.670
Q21_S	0.344
Q22_S	0.559
Q8_A	0.260
Q9_A	0.395
Q12_A	0.511
Q13_A	0.439
Q14_A	0.535
Q23_A	0.337
Q24_A	0.242
Q25_A	0.293

The R² values for observed indicators remain nearly identical to the original four-factor model, as expected since item-factor relationships are unaffected by removing Q5P from the psychological domain. Items maintain their explained variance patterns: Q17F (physical; R² = 0.851), Q18F (physical; R² = 0.788), Q19P (psycho; R² = 0.755), and Q10F (physical; R² = 0.657) exhibit the highest R² values. Weaker indicators persist: Q4F (physical; R² = 0.179), Q24A (environment; R² = 0.242), and Q3F (physical; R² = 0.260). The psychological domain now comprises only five items (Q6P through Q26P, excluding Q5P), with R² values ranging from 0.420 (Q26P) to 0.755 (Q19P). The absence of Q5P does not substantially alter the proportion of variance explained in remaining psychological items, confirming that its removal primarily addresses construct validity concerns (cross-loadings, discriminant validity) rather than indicator quality within its domain. Researchers should note that while Q5P contributed to psychological domain reliability (ω decreased from 0.823 to 0.796), its exclusion may strengthen theoretical coherence by reducing multidimensional content.

Parameter estimates

Factor loadings

Factor	Indicator	Symbol	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
							Lower	Upper	
physical	Q3_F	λ11	0.510	0.026	19.526	< .001	0.459	0.562	0.510
	Q4_F	λ12	0.423	0.029	14.737	< .001	0.367	0.479	0.423
	Q10_F	λ13	0.811	0.012	66.021	< .001	0.787	0.835	0.811
	Q15_F	λ14	0.643	0.025	25.889	< .001	0.595	0.692	0.643
	Q16_F	λ15	0.590	0.023	25.647	< .001	0.545	0.635	0.590
	Q17_F	λ16	0.923	0.007	125.428	< .001	0.908	0.937	0.923
psycho	Q18_F	λ17	0.888	0.009	103.944	< .001	0.871	0.905	0.888
	Q6_P	λ21	0.659	0.019	34.689	< .001	0.622	0.697	0.659
	Q7_P	λ22	0.677	0.018	37.254	< .001	0.642	0.713	0.677
	Q11_P	λ23	0.671	0.019	34.426	< .001	0.633	0.709	0.671
	Q19_P	λ24	0.869	0.012	71.132	< .001	0.845	0.893	0.869
	Q26_P	λ25	0.648	0.020	32.730	< .001	0.609	0.687	0.648
social	Q20_S	λ31	0.819	0.019	43.875	< .001	0.782	0.855	0.819
	Q21_S	λ32	0.586	0.027	21.558	< .001	0.533	0.640	0.586
environment	Q22_S	λ33	0.748	0.020	37.362	< .001	0.709	0.787	0.748
	Q8_A	λ41	0.510	0.028	17.913	< .001	0.454	0.566	0.510
	Q9_A	λ42	0.629	0.025	25.608	< .001	0.581	0.677	0.629
	Q12_A	λ43	0.715	0.021	34.196	< .001	0.674	0.756	0.715
	Q13_A	λ44	0.663	0.023	28.432	< .001	0.617	0.708	0.663
	Q14_A	λ45	0.732	0.021	35.663	< .001	0.691	0.772	0.732
	Q23_A	λ46	0.580	0.027	21.770	< .001	0.528	0.633	0.580
	Q24_A	λ47	0.492	0.028	17.504	< .001	0.437	0.547	0.492
	Q25_A	λ48	0.542	0.027	19.777	< .001	0.488	0.595	0.542

All standardized factor loadings remain statistically significant ($p < .001$) and exhibit patterns nearly identical to the original four-factor model. Physical domain loadings range from 0.423 (Q4F) to 0.923 (Q17F), psychological domain loadings (now five items) range from 0.648 (Q26P) to 0.869 (Q19P), social domain from 0.586 (Q21S) to 0.819 (Q20S), and environment domain from 0.492 (Q24A) to 0.732 (Q14A). The psychological domain retains strong measurement properties despite losing Q5P: all five remaining items exceed the 0.60 threshold, with four items surpassing 0.70 (Q6P = 0.659, Q7P = 0.677, Q11P = 0.671, Q19P = 0.869). Item Q5P originally had a loading of 0.740 in the full model, indicating it was a strong indicator of psychological quality of life. However, its extensive cross-loadings with other domains (particularly environment and physical) suggested it measured broader wellbeing beyond the specific psychological facet. The removal improves discriminant validity while maintaining adequate domain representation through the remaining five psychological items.

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
physical	1.000	0.000			1.000	1.000	1.000
psycho	1.000	0.000			1.000	1.000	1.000
social	1.000	0.000			1.000	1.000	1.000
environment	1.000	0.000			1.000	1.000	1.000

Factor Covariances

							95% Confidence Interval		
			Estimate	Std. Error	z-value	p	Lower	Upper	Std. Est. (all)
physical	↔	psycho	0.922	0.010	96.703	< .001	0.903	0.941	0.922
physical	↔	social	0.638	0.024	26.528	< .001	0.591	0.685	0.638
physical	↔	environment	0.630	0.022	28.640	< .001	0.587	0.673	0.630
psycho	↔	social	0.755	0.019	39.028	< .001	0.717	0.793	0.755
psycho	↔	environment	0.659	0.024	27.049	< .001	0.611	0.706	0.659
social	↔	environment	0.550	0.028	19.517	< .001	0.495	0.605	0.550

The latent factor correlations in the model without Q5P show notable differences from the original four-factor model. Most strikingly, the physical-psycho correlation ($r = 0.922$) remains extremely high, approaching multicollinearity concerns and actually slightly increasing from the original model ($r = 0.909$). This persistent high correlation suggests that removing Q5P did not resolve the empirical overlap between physical and psychological quality of life domains. Other correlations include psycho-social ($r = 0.755$, decreased from 0.771), psycho-environment ($r = 0.659$, decreased from 0.725), physical-social ($r = 0.638$, unchanged), physical-environment ($r = 0.630$, unchanged), and social-environment ($r = 0.550$, unchanged). The reduction in psycho-environment correlation (from 0.725 to 0.659) suggests Q5P contributed substantially to the overlap between these domains, supporting its removal to improve discriminant validity. However, the persistently high physical-psycho correlation (> 0.85) continues to justify testing alternative structures such as bifactor or second-order models, which may better represent a general quality of life factor underlying these highly correlated domains.

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
Q3_F	0.740	0.000			0.740	0.740	0.740
Q4_F	0.821	0.000			0.821	0.821	0.821
Q10_F	0.343	0.000			0.343	0.343	0.343
Q15_F	0.586	0.000			0.586	0.586	0.586
Q16_F	0.652	0.000			0.652	0.652	0.652
Q17_F	0.149	0.000			0.149	0.149	0.149
Q18_F	0.212	0.000			0.212	0.212	0.212
Q6_P	0.565	0.000			0.565	0.565	0.565
Q7_P	0.541	0.000			0.541	0.541	0.541
Q11_P	0.550	0.000			0.550	0.550	0.550
Q19_P	0.245	0.000			0.245	0.245	0.245
Q26_P	0.580	0.000			0.580	0.580	0.580
Q20_S	0.330	0.000			0.330	0.330	0.330
Q21_S	0.656	0.000			0.656	0.656	0.656
Q22_S	0.441	0.000			0.441	0.441	0.441
Q8_A	0.740	0.000			0.740	0.740	0.740
Q9_A	0.605	0.000			0.605	0.605	0.605
Q12_A	0.489	0.000			0.489	0.489	0.489
Q13_A	0.561	0.000			0.561	0.561	0.561
Q14_A	0.465	0.000			0.465	0.465	0.465
Q23_A	0.663	0.000			0.663	0.663	0.663
Q24_A	0.758	0.000			0.758	0.758	0.758
Q25_A	0.707	0.000			0.707	0.707	0.707

Residual covariances

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
Q3_F	↔	Q4_F	0.387	0.025	15.605	< .001	0.338	0.435

Modification Indices

Cross-loadings

			Mod. Ind.	EPC
environment	→	Q15_F	94.407	0.351
physical	→	Q24_A	78.505	−0.281
psycho	→	Q24_A	77.213	−0.289
environment	→	Q17_F	43.449	−0.232
environment	→	Q18_F	42.877	−0.228
psycho	→	Q10_F	42.388	0.704
physical	→	Q6_P	36.087	−0.700
social	→	Q24_A	33.243	−0.203
physical	→	Q21_S	31.008	0.235
psycho	→	Q21_S	26.547	0.277
social	→	Q6_P	18.702	0.220
environment	→	Q10_F	17.672	0.138
psycho	→	Q22_S	17.399	−0.322
physical	→	Q22_S	16.388	−0.233
psycho	→	Q17_F	16.363	−0.483
social	→	Q7_P	12.950	−0.202
social	→	Q16_F	12.632	0.158
social	→	Q10_F	12.223	0.149
physical	→	Q14_A	12.052	0.111
psycho	→	Q14_A	10.753	0.110

Modification indices for cross-loadings show substantial reductions compared to the original four-factor model, confirming that Q5P removal addressed major misspecification sources. The highest MI value (environment → Q15F, MI = 94.407, EPC = 0.351) is considerably lower than the original model's highest MI (environment → Q5P, MI = 175.788). Other notable cross-loadings include physical/psycho → Q24A (MI > 77), psycho → Q10F (MI = 42.388), and physical → Q6P (MI = 36.087). These remaining cross-loadings are substantially smaller than those involving Q5P in the original model, indicating improved model specification. The lower MI values overall suggest that the four-factor structure without Q5P better accommodates the covariance patterns in the data. However, some cross-loadings persist, particularly between physical and psychological domains (Q6P, Q10F), reflecting the continued high correlation (r = 0.922) between these factors. While introducing these cross-loadings could marginally improve fit, doing so would compromise parsimony and interpretability, particularly given that the current model already demonstrates acceptable fit.

Residual covariances

			Mod. Ind.	EPC
Q24_A	↔	Q25_A	114.934	0.273
Q17_F	↔	Q18_F	84.659	0.157
Q10_F	↔	Q18_F	31.926	−0.118
Q3_F	↔	Q15_F	31.211	0.178
Q23_A	↔	Q25_A	23.889	0.140
Q15_F	↔	Q13_A	21.582	0.146
Q4_F	↔	Q15_F	21.431	0.147
Q15_F	↔	Q14_A	20.568	0.143
Q22_S	↔	Q23_A	18.885	0.131
Q15_F	↔	Q12_A	17.083	0.134
Q10_F	↔	Q11_P	16.824	0.098
Q15_F	↔	Q25_A	16.632	0.121
Q23_A	↔	Q24_A	15.709	0.117
Q14_A	↔	Q25_A	15.573	−0.128
Q10_F	↔	Q26_P	15.100	0.102
Q16_F	↔	Q21_S	14.711	0.117
Q8_A	↔	Q9_A	14.662	0.107
Q20_S	↔	Q22_S	14.362	0.132
Q15_F	↔	Q19_P	13.578	−0.123
Q15_F	↔	Q17_F	12.722	−0.110
Q10_F	↔	Q14_A	12.198	0.094
Q7_P	↔	Q8_A	11.854	0.099
Q3_F	↔	Q6_P	11.351	−0.112
Q3_F	↔	Q19_P	11.301	−0.114
Q10_F	↔	Q17_F	11.297	−0.066
Q7_P	↔	Q24_A	10.725	−0.113

Residual covariances indicate local dependencies between item pairs beyond what the common factors explain. The highest MI values occur for Q24A-Q25A (MI = 114.934, EPC = 0.273), Q17F-Q18F (MI = 84.659, EPC = 0.157), Q10F-Q18F (MI = 31.926), and Q3F-Q15F (MI = 31.211). The Q24A-Q25A pair (both environment items about physical safety and home environment) shows increased MI compared to the original model, potentially reflecting shared method variance or semantic similarity. The Q17F-Q18F residual correlation (pain and medical treatment items) remains consistently high across all model specifications, suggesting these items share unique variance beyond the physical factor. The Q3F-Q4F residual covariance was freely estimated in the model (estimate = 0.387, p < .001), acknowledging these reverse-coded items' unique relationship. Overall, residual covariances show similar patterns to the original model, indicating Q5P removal primarily impacted factor-level relationships (correlations, cross-loadings) rather than item-specific local dependencies. Freeing additional residual correlations (e.g., Q24A-Q25A) would improve fit but requires theoretical justification to avoid capitalization on chance.

Plots

Model evaluation and comparison: The four-factor model excluding Q5P demonstrates marginal fit improvements over the original model (CFI increased from 0.985 to 0.988, RMSEA decreased from 0.065 to 0.057) while substantially reducing problematic cross-loadings. The psychological domain retains adequate reliability ($\omega = 0.796$) and strong factor loadings despite losing one item. The primary benefit of Q5P removal is enhanced discriminant validity, as evidenced by the reduction in psycho-environment correlation (from 0.725 to 0.659) and dramatically lower modification indices for cross-loadings. However, the persistently high physical-psycho correlation ($r = 0.922$) suggests continued empirical overlap between these domains, justifying comparison with bifactor or second-order structures. From a practical standpoint, researchers must weigh the trade-off between (1) retaining Q5P for slightly higher reliability and complete domain coverage versus (2) removing Q5P for improved construct distinctiveness and reduced multidimensionality. This decision should be guided by the study's theoretical framework and whether discriminant validity between psychological and other domains is critical for subsequent structural analyses. The model comparison section provides formal statistical tests to evaluate whether Q5P removal significantly degrades fit.

Misfit plot

	Q3_F	Q4_F	Q10_F	Q15_F	Q16_F	Q17_F	Q18_F	Q6_P	Q7_P	Q11_P	Q19_P	Q26_P	Q20_S	Q21_S	Q22_S	Q8_A	Q9_A	Q12_A	Q13_A	Q14_A	Q23_A	Q24_A	Q25_A
Q3_F	0																						
Q4_F	0	0																					
Q10_F	.05	.02	0																				
Q15_F	.16	.13	.03	0																			
Q16_F	.03	0	.01	.09	0																		
Q17_F	.02	.02	.04	.09	.01	0																	
Q18_F	.03	.03	.08	.06	.04	.04	0																
Q6_P	.1	.07	.02	.07	.03	.06	.05	0															
Q7_P	.04	.02	.05	.05	.02	0	.04	.02	0														
Q11_P	.05	.07	.08	.03	.06	.06	.06	0	.01	0													
Q19_P	.1	.08	.01	.11	.03	.02	.03	.02	.03	.03	0												
Q26_P	.03	.03	.09	.07	.06	.01	.05	.07	0	.04	0	0											
Q20_S	.07	.09	.01	.06	.03	.01	.01	.06	.03	.03	.03	.03	0										
Q21_S	.04	.02	.06	0	.11	.04	.05	.02	.05	.01	.02	0	.06	0									
Q22_S	.03	.05	.02	.04	.02	.04	.01	.04	.08	.01	0	.06	.03	.04	0								
Q8_A	.03	.01	.05	.07	.01	.03	.02	.03	.09	.04	0	.02	0	.05	.04	0							
Q9_A	.03	.04	.08	.08	.02	.05	.02	.02	.06	.05	.02	.01	.03	.09	.01	.09	0						
Q12_A	.03	.07	.06	.13	.01	.06	.06	.05	.01	.06	.01	.03	.07	0	.03	.08	.08	0					
Q13_A	.03	.03	.01	.14	0	.04	.06	.02	.05	.04	.04	0	.01	.08	.03	.08	.06	.06	0				
Q14_A	.03	.05	.08	.13	.06	.03	.08	.04	.02	.02	0	.01	.01	.01	.04	.06	.07	.04	.03	0			
Q23_A	.05	.04	.01	.06	.01	.09	.08	.01	.04	0	.01	.01	.05	.01	.12	.04	.06	.07	.07	.06	0		
Q24_A	0	.06	.09	.02	.04	.09	.09	.03	.11	.06	.09	.09	.02	0	.04	.01	0	.02	.03	.04	.1	0	
Q25_A	.03	.01	.07	.11	.03	.06	.04	.04	.02	.01	.04	.07	.01	.01	.01	.01	.03	.05	.05	.11	.12	.23	0

Models comparison

Model fit

	AIC	BIC	n	Baseline test			Difference test		
				χ^2	df	p	$\Delta\chi^2$	Δdf	p
4F-bifactor			1047	583.481	197	< .001			
4F-B			1047	977.820	223	< .001	394.339	26	< .001
4F-2nd			1047	1007.178	226	< .001	29.358	3	< .001
4F-A			1047	1339.798	245	< .001	332.620	19	< .001

Note. The AIC, BIC and additional information criteria are only available with ML-type estimators
Note. some models are based on a different set of observed variables

The baseline test presents chi-square values for each model: 4F-bifactor ($\chi^2 = 583.481$, $df = 197$), 4F-B [4F without Q5_F] ($\chi^2 = 977.820$, $df = 223$), 4F-2nd ($\chi^2 = 1007.178$, $df = 226$), and 4F-A [original 4F] ($\chi^2 = 1339.798$, $df = 245$). All models show significant chi-square tests ($p < .001$), but as consistently emphasized, this reflects sample size sensitivity ($n = 1047$) rather than poor fit. The difference test provides chi-square difference tests between nested models. Comparing 4F-B against 4F-bifactor yields $\Delta\chi^2 = 394.339$ ($\Delta df = 26$, $p < .001$), indicating statistically significant fit improvement for the bifactor model. Similarly, 4F-2nd versus 4F-bifactor shows $\Delta\chi^2 = 29.358$ ($\Delta df = 3$, $p < .001$), and 4F-A versus 4F-bifactor shows $\Delta\chi^2 = 332.620$ ($\Delta df = 19$, $p < .001$). These significant chi-square difference tests favor the bifactor model over all alternatives. However, **chi-square difference tests inherit the same sample size sensitivity as absolute chi-square tests**, often rejecting practically equivalent models in large samples. Researchers should not rely solely on these tests but consider effect size measures of fit improvement alongside theoretical interpretability and parsimony.

Additional fit measures

Fit indices

Index	4F-A	4F-B	4F-bifactor	4F-2nd
Comparative Fit Index (CFI)	0.985	0.988	0.994	0.988
T-size CFI	0.982	0.986	0.992	0.986
Tucker-Lewis Index (TLI)	0.983	0.987	0.992	0.986
Bentler-Bonett Non-normed Fit Index (NNFI)	0.983	0.987	0.992	0.986
Bentler-Bonett Normed Fit Index (NFI)	0.981	0.985	0.991	0.984
Parsimony Normed Fit Index (PNFI)	0.871	0.868	0.772	0.879
Bollen's Relative Fit Index (RFI)	0.979	0.983	0.988	0.982
Bollen's Incremental Fit Index (IFI)	0.985	0.988	0.994	0.988
Relative Noncentrality Index (RNI)	0.985	0.988	0.994	0.988

Note. T-size CFI is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: 4F-2nd
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: 4F-A
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: 4F-B
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.885 < fair < 0.94 < close** for model: 4F-bifactor

The comparative fit indices reveal systematic patterns across models. **CFI values:** 4F-bifactor (0.994) > 4F-B (0.988) $\approx$ 4F-2nd (0.988) > 4F-A (0.985), with T-size CFI showing identical rankings. **TLI/NNFI values:** 4F-bifactor (0.992) > 4F-B (0.987) $\approx$ 4F-2nd (0.986) > 4F-A (0.983). **RMSEA values:** 4F-bifactor (0.043, 90% CI [0.039, 0.047]) < 4F-B (0.057) = 4F-2nd (0.057) < 4F-A (0.065), with the bifactor model showing the lowest misfit. **SRMR values:** 4F-bifactor (0.041) < 4F-B (0.053) < 4F-2nd (0.054) < 4F-A (0.057). These indices consistently favor the bifactor model, followed by the four-factor model without Q5 and the second-order model (nearly tied), with the original four-factor model showing the poorest fit. However, **critical caution is warranted:** as discussed in Appendix Table 3, traditional cutoff values (CFI > .95, RMSEA < .06) should not be applied mechanically. Bifactor models are structurally more complex, inherently producing better fit indices due to additional parameters, which can artificially inflate fit without necessarily reflecting superior theoretical validity. The fit improvements, while statistically significant, represent relatively small effect size differences (e.g., Δ CFI = 0.009 between 4F-A and bifactor). **Model selection should prioritize theoretical interpretability, practical utility, and parameter stability** over marginal fit improvements. The T-size indices (T-size CFI, T-size RMSEA) adjust for model complexity and sample size, providing alternative perspectives that still favor the bifactor model but with reduced magnitude.

Other fit measures

Metric	4F-A	4F-B	4F-bifactor	4F-2nd
Root mean square error of approximation (RMSEA)	0.065	0.057	0.043	0.057
RMSEA 90% CI lower bound	0.062	0.053	0.039	0.054
RMSEA 90% CI upper bound	0.069	0.061	0.047	0.061
RMSEA p-value	1.537 $\times 10^{-13}$	9.592 $\times 10^{-4}$	0.997	3.411 $\times 10^{-4}$
T-size RMSEA	0.069	0.061	0.047	0.061
Standardized root mean square residual (SRMR)	0.057	0.053	0.041	0.054
Hoelter's critical N ($\alpha = .05$)	221.561	277.884	414.656	273.171
Hoelter's critical N ($\alpha = .01$)	234.759	295.227	442.171	290.109
Goodness of fit index (GFI)	0.987	0.989	0.994	0.989
McDonald fit index (MFI)	0.593	0.697	0.831	0.688
Expected cross validation index (ECVI)				

Note. T-size RMSEA is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.087 < poor** for model: 4F-bifactor
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: 4F-2nd
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: 4F-A
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: 4F-B

The detailed RMSEA metrics reinforce the fit patterns. The bifactor model demonstrates close fit (RMSEA = 0.043, 90% CI [0.039, 0.047], p-value = 0.997), with the p-value testing the null hypothesis that RMSEA $\leq$ 0.05 (close fit). The high p-value (0.997) suggests the data strongly support close fit for the bifactor model. The T-size RMSEA (0.047) adjusts for sample size and model complexity, remaining below 0.05. Models 4F-B and 4F-2nd show acceptable fit (RMSEA = 0.057, T-size RMSEA = 0.061), with 90% CI upper bounds at 0.061, marginally exceeding the "close fit" threshold but well below "poor fit" (> 0.08). The original 4F-A model shows adequate but weaker fit (RMSEA = 0.065, T-size RMSEA = 0.069, 90% CI upper bound = 0.069). Hoelter's critical N values indicate the sample size at which the chi-square test would not reject the model at $\alpha = .05$ and $\alpha = .01$. The bifactor model shows the highest critical N (.05 = 414.656, .01 = 442.171), indicating it would maintain acceptable fit in much smaller samples, while 4F-A shows the lowest (221.561, 234.759), suggesting it requires the current large sample to avoid rejection. However, **RMSEA cutoffs must be interpreted contextually:** these were derived from specific simulation conditions (ML estimation, continuous data, three factors, moderate loadings) that differ from the current DWLS estimation with ordinal data and four factors. Dynamic Fit Index (DFI) cutoffs specific to each model's characteristics provide more appropriate benchmarks (see supplementary materials).

	R²			
	4F-A	4F-B	4F-bifactor	4F-2nd
Q3_F	0.258	0.260	0.577	0.260
Q4_F	0.179	0.179	0.471	0.178
Q10_F	0.667	0.657	0.678	0.657
Q15_F	0.409	0.414	0.515	0.413
Q16_F	0.351	0.348	0.353	0.348
Q17_F	0.849	0.851	0.865	0.852
Q18_F	0.786	0.788	0.811	0.789
Q5_P	0.548			
Q6_P	0.449	0.435	0.469	0.436
Q7_P	0.441	0.459	0.451	0.460
Q11_P	0.434	0.450	0.445	0.450
Q19_P	0.740	0.755	0.804	0.761
Q26_P	0.410	0.420	0.413	0.420
Q20_S	0.668	0.670	0.726	0.670
Q21_S	0.340	0.344	0.277	0.346
Q22_S	0.564	0.559	0.573	0.558
Q8_A	0.260	0.260	0.231	0.261
Q9_A	0.390	0.395	0.358	0.397
Q12_A	0.508	0.511	0.475	0.511
Q13_A	0.422	0.439	0.417	0.439
Q14_A	0.586	0.535	0.477	0.535
Q23_A	0.325	0.337	0.395	0.336
Q24_A	0.232	0.242	0.432	0.242
Q25_A	0.280	0.293	0.370	0.293
psycho				1.000
physical				0.832
social				0.548
environment				0.469

The R² comparative table reveals how different model structures partition variance in observed indicators. **Item-level patterns:** Most items show relatively stable R² across models 4F-A, 4F-B, and 4F-2nd (e.g., Q17F ranges 0.849-0.852, Q18F ranges 0.786-0.789), as expected since these models specify similar first-order factor structures. The bifactor model shows markedly different R² values due to its dual-loading structure, with some items showing substantial increases (Q3F: 0.258 in 4F-A → 0.577 in bifactor; Q4F: 0.179 → 0.471; Q20S: 0.668 → 0.726) and others showing modest increases or decreases. **Absence of Q5P:** The 4F-A model includes Q5P (R² = 0.548), while all other models exclude it, making direct comparison incomplete. **First-order factor R² in 4F-2nd:** The second-order model shows R² for first-order factors (psycho = 1.000, physical = 0.832, social = 0.548, environment = 0.469), indicating variance explained by the higher-order QOL factor. The perfect R² = 1.000 for psycho reflects the identification constraint (variance fixed to zero), not empirical redundancy. **Interpretation:** Higher R² values in the bifactor model do not automatically indicate superior measurement - they reflect the additional general factor capturing common variance. Researchers should evaluate whether the R² increases represent meaningful common variance (justifying a general factor) or statistical artifact from model complexity.

Parameter estimates

Comprehensive model comparison and selection guidance: The four models represent theoretically distinct structures for quality of life measurement: (1) **4F-A** (original four correlated factors) assumes distinct but related domains; (2) **4F-B** (four factors without Q5P) addresses discriminant validity concerns; (3) **4F-2nd** (second-order) posits domains as manifestations of general QOL; (4) **4F-bifactor** allows items to load on both general QOL and specific domains simultaneously. Statistical criteria consistently favor the bifactor model (best fit indices), followed by 4F-B and 4F-2nd (nearly equivalent), with 4F-A showing weakest fit. However, **fit indices alone should not dictate model selection**. Critical considerations include: **(1) Theoretical coherence:** Does the chosen structure align with quality of life theory and prior research? Systematic reviews support the four-correlated-factor structure most strongly. **(2) Practical utility:** Will domain-specific scores be used? If only a total score is needed, bifactor or second-order models are justified; if domain profiles are interpreted, the four-factor correlated structure is more appropriate. **(3) Parameter stability:** Bifactor models are prone to overfitting and may show unstable parameters across samples - replication is essential. **(4) Parsimony:** Simpler models (4F-A, 4F-B) are easier to interpret and replicate, while complex models (bifactor) require larger samples and may not generalize. **(5) Discriminant validity:** The persistently high physical-psycho correlation (r > .90 in 4F models) suggests empirical overlap, supporting hierarchical structures (bifactor, second-order) that acknowledge this commonality. **Recommendation:** Researchers should report multiple models, evaluate bifactor reliability indices (ωH, ωS, ECV) using lavaan or specialized packages (not available in JASP 0.17.1), and justify their final choice based on theoretical and practical considerations rather than fit indices alone. If domain-specific interpretations are critical and theoretically justified despite high intercorrelations, retain the four-factor structure (preferably 4F-B for improved discriminant validity). If a general quality of life construct is theoretically paramount, the bifactor model offers the most flexible representation, but only if reliability analyses support meaningful specific factors beyond the general factor.

4F-A
4F-B
4F-bifactor
4F-2nd

Covariance tables

4F-A
4F-B
4F-bifactor
4F-2nd

